Received: 20 November 2019 | Revised: 13 February 2020 | Accepted: 10 April 2020

DOI: 10.1111/lnc3.12372

ARTICLE

WILEY

Sound symbolism and theoretical phonology

Shigeto Kawahara

Institute of Cultural and Linguistics Studies, Keio University, Tokyo, Japan

Correspondence
Shigeto Kawahara, Institute of Cultural and Linguistic Studies, Keio University, Tokyo, Japan.
Email: kawahara@icl.keio.ac.jp

Funding information
Japan Society for the Promotion of Science, Grant/Award Number: #17K13448

Abstract

A received wisdom in modern linguistic theories is that the relationships between sounds and meanings are generally arbitrary. However, there is a growing body of evidence suggesting that in some cases sounds and meanings have systematic relationships—patterns known as "sound symbolism." Yet most of these studies are conducted by psychologists, cognitive scientists, or cognitive linguists, and currently, only a few theoretical phonologists pay serious attention to sound symbolism. This paper reviews major studies on sound symbolism and argues that sound symbolism can be an interesting topic of exploration for theoretical phonologists. This paper also demonstrates that insights gained by phonological research can shed light on some important issues in the studies of sound symbolism, and vice versa. Overall, I hope that this paper is informative for both theoretical phonologists and researchers who work on sound symbolism.

1 | INTRODUCTION

One of the standard assumptions often taken for granted in modern linguistic theories is the arbitrariness of signs—the relationship between sounds and meaning is arbitrary. While this thesis was already known since the time of Plato's *Cratylus*, Saussure (1916) perhaps played a key role in establishing this thesis at the center of modern linguistic theories. Also influential was Hockett (1959), who claimed that the arbitrariness of signs is one design feature that distinguishes human languages from other animal communication systems. Language is undoubtedly a system that is capable of associating sounds and meanings in arbitrary ways, which allows it to have strong expressive power (Lupyan & Winter, 2018). Nevertheless, there are cases in which systematic associations between sounds and meanings hold; these patterns are known as "sound symbolism." One well-known example is the observation that speakers of many languages feel nonce words containing [a] (e.g., [mal]) to be larger than those containing [i] (e.g., [mil]) (Berlin, 2006; Newman, 1933; Sapir, 1929; Shinohara & Kawahara, 2016; Ultan, 1978). Another well-known case is the *takete-maluma* effect (Köhler, 1947), in which names with voiceless obstruents tend to be associated with angular shapes, while names with sonorants tend to be associated with round shapes (Sidhu, Deschamps, Bourdage, & Pexman, 2019). These sound symbolic effects are observed in experimental settings as well as in the form of statistical skews in the lexicon (see the overview papers cited below).

The rise of interests in sound symbolism is partly evidenced by the number of recent overview articles on sound symbolism, each written from a slightly different perspective (Akita, 2015; Cuskley & Kirby, 2013; Dingemanse, Blasi, Lupyan, Christiansen, & Monaghan, 2015; Lockwood & Dingemanse, 2015; Nuckolls, 1999; Perniss, Thompson, & Vigiliocco, 2010; Schmidtke, Conrad, & Jacobs, 2014; Sidhu & Pexman, 2018; Spence, 2011; Svantesson, 2017). Given that so many overview papers already exist, one may wonder if there is a need for another overview paper. I contend that the answer is positive, because none of these papers are directed at theoretical phonologists. Nevertheless, I believe that studying sound symbolism can offer important insights into the architecture of phonological knowledge, and also that phonological studies have much to offer for the studies of sound symbolism.

It is helpful to start this discussion by considering why sound symbolism has generally been considered as residing outside the realm of theoretical phonology. To the extent that phonological knowledge is about what speakers know about the sound structure of their native language, if systematic connections between sounds and meanings exist, then, exclusion of sound symbolism from a topic of phonological inquiry does not seem to be clearly motivated.[1] But then why is sound symbolism not actively studied in theoretical phonology?[2] Of course, the influences by Saussure and Hockett must have been non-negligible. Another possible reason may be that generative linguistic theories do not, or did not, accept probabilistic tendencies as belonging to competence—it is, or was, believed that grammars should only make a dichotomous, grammatical vs. ungrammatical distinction (Schütze, 1996). However, we now have an extended body of evidence showing that phonological knowledge can be stochastic (Boersma & Hayes, 2001; Hayes & Londe, 2006; Pierrehumbert, 2001; Zuraw & Hayes, 2017), and several grammatical models have made it possible to model stochastic phonological knowledge (Coetzee & Pater, 2011). This rise of new stochastic grammatical models removed one obstacle that prevented us from exploring stochastic sound symbolic patterns (Section 2.3).

2 | PHONOLOGY AND SOUND SYMBOLISM

Some theoretical phonologists have started analyzing sound symbolism—in particular, alternation patterns that are caused by sound symbolic principles—as a part of their phonological inquiry. This section reviews some of these studies and presents other arguments that studies of sound symbolism are not as irrelevant as broadly assumed in theoretical phonology.

2.1 | Some alternations are motivated by sound symbolic principles

The first reason to study sound symbolism from the perspective of theoretical phonology is the emerging observation that some alternations appear to be motivated by sound symbolic considerations. In particular, Alderete and Kochetov (2017) have shown that, for example, palatalization in Japanese baby talk (what they call "expressive palatalization") shows a number of properties that are different from purely phonological palatalization processes. First, phonological palatalization is assimilatory in nature, usually caused by (high) front vowels or palatal glides; expressive palatalization in Japanese, however, does not require such a trigger, changing all /s/ in a target word into either [ʃ] or [tʃ] (e.g., /o**s**akana-**s**aN/ → [o**ʃ**akana-**ʃ**aɴ] or [o**tʃ**akana-**tʃ**aɴ] "fish(−*y*)"). Second, affrication of fricatives can occur in expressive palatalization, as in the previous example, but never occurs in purely phonological palatalization (Bhat, 1978). Third, expressive palatalization shows place and manner asymmetries that are not shared by purely phonological palatalization; for instance, the majority of expressive palatalization patterns targets coronal consonants, and moreover, when non-coronals are targeted, so are coronals. Fourth, expressive palatalization characteristically targets all relevant consonants within a word, which is rare at best for purely phonological processes. Finally, expressive palatalization can create an otherwise phonotactically illegal sequence; for example, /**s**en**s**e:/"teacher" can become [**tʃ**en**tʃ**e:], but [tʃe] is not a legal CV combination (Ito & Mester, 1995). Expressive palatalization thus does not seem to be motivated by purely phonological considerations; instead, they appear to occur to express sound symbolic meanings.

These alternations driven by sound symbolic principles can interact with other phonological considerations within a single grammatical system. To illustrate, Kumagai (2019) shows that in Japanese nickname formation, /h/ can be turned into [p] (e.g., /**h**aruka/→ [**p**aruru]). It is hard to consider this alternation to be caused by a phonological constraint, since [p] is at best a marked segment in the native phonology of Japanese (Ito & Mester, 1995). Kumagai argues that this alternation is instead caused by a sound symbolic principle to express cuteness; indeed his experiment shows that [*p*] is judged to be "cuter" than any other consonants. Moreover, he has shown that this alternation interacts with an independently motivated phonotactic constraint that prohibits the [p…[+voice]] configuration (Kawahara, 2018)—nicknames created by the [h]-[p] alternation are judged to be less natural when they violate *[p…[+voice]]. See Jang (2019) for a similar case found in Korean baby talk, *Aegyo*. These observations have led these authors to propose that sound symbolic principles should be integrated with "core phonological grammar."

2.2 | The role of distinctive features in sound symbolism

Another aspect of sound symbolism that makes it interesting for theoretical phonologists is the observation that some sound symbolic patterns operate at the level of distinctive features rather than individual segments. For example, secondary palatalization in Japanese mimetic words denotes, among other expressive meanings, "uncontrolledness" (Hamano, 1998). This pattern should be characterized using a distinctive feature (e.g., [palatal] or [−back]) because consonants at all places of articulation can be affected by it (although there is a preference toward targeting coronal consonants). Indeed, this palatalization pattern, despite its sound symbolic nature, has been analyzed by a number of theoretical phonologists (see Alderete & Kochetov, 2017: 732).

To take an example from a sound–meaning association, Kumagai and Kawahara (2020) point out that almost all diaper names in Japanese contain [p] and/or [m] (e.g., [**m**a**m**ii**p**oko]); when asked to produce new diaper names, Japanese speakers tend to use *all* types of labial consonants ([p], [b], [ɸ], [m], and [w]) more often than when they are asked to come up with new names for adult cosmetics. This result suggests that the sound symbolism at issue operates at the level of a distinctive feature—[labial]—rather than at the level of individual segments. Considering that the existing diaper names contain only [p] and [m] but not other labial consonants, the participants of their experiment seem to have shown feature-based generalizations, just as "normal" phonological patterns do (Albright, 2009; Finley & Bedecker, 2009).[3] This observation raises the possibility that phonological patterns and sound symbolic patterns can use the same set of vocabularies, that is, distinctive features. See also D'Onofrio (2014), Jakobson (1960), McCarthy (1983), and Nobile (2015) for related discussion.

2.3 | Modeling sound symbolism with formal grammatical frameworks

Mainstream generative models of grammar have long considered sound symbolism as residing outside the purview of linguistic knowledge proper. However, the existence of sound symbolism may require us to posit a mechanism that directly connects sounds and meanings, and some studies propose to use a formal grammatical model to capture such connections. For instance, as discussed above, Alderete and Kochetov (2017) demonstrate that expressive palatalization patterns are motivated by sound symbolic principles. They propose an analysis of expressive palatalization using Optimality Theory (Prince & Smolensky, 2004), with a set of violable constraints (Express[x]) specifying which sounds should be realized to express which meanings. Kawahara, Katsuda, and Kumagai (2019) analyze sound symbolic connections themselves (rather than alternations triggered by sound symbolic considerations), and argue that as long as generative phonology is a function that maps one representation (e.g., underlying forms) to another (e.g., surface forms)—as it in fact has been—there is nothing that prevents us from using the same formalism to model the mapping from representation in one modality (i.e., sound) to representation in another modality (i.e., meaning). In other words, the grammatical device that phonologists have been using for decades can be applied to formalize sound symbolic connections at no additional costs.

As stated in Section 1, sound symbolic connections are almost always stochastic—certain sounds *tend* to be associated with certain meanings, but such associations are never deterministic (Dingemanse, 2018). Likewise, recent studies have demonstrated that phonological knowledge can be stochastic; for example, some structures tend to be preferred over others, and some alternations are more likely to occur in one environment than in other environments (Boersma & Hayes, 2001; Hayes & Londe, 2006; Pierrehumbert, 2001; Zuraw & Hayes, 2017). Focusing on this parallel between phonological patterns and sound symbolic connections, Kawahara et al. (2019) propose to use the MaxEnt model with Optimality Theoretic constraints (Goldwater & Johnson, 2003) to account for stochastic aspects of sound symbolic patterns. For example, in Japanese Takarazuka actress names, the more sonorants are contained in their name, the more likely that name is used for a female name. MaxEnt grammar with a constraint like *SonorantInMaleName accounts for the stochastic nature of this sound symbolic pattern—this analysis is fundamentally the same as a MaxEnt analysis of "normal" phonological patterns (e.g., Zuraw & Hayes, 2017). In short, phonological patterns and sound symbolic patterns share a non-trivial nature (stochasticity), and an analytical tool like MaxEnt is able to capture that parallel.

3 | COMMON ISSUES AND SHARED INTERESTS

There are common issues that are addressed both by theoretical phonologists and those who study sound symbolism. My intention in this section is not to solve any of these issues, but to show that we have shared interests. By making these shared interests clear, it is hoped that insights gained in one domain of inquiry can shed light on questions that are addressed in the other. I also hope that these parallels pique phonologists' interests to study more about sound symbolism.

3.1 | Phonetic naturalness

One continuing debate in phonological theory is to what extent phonological patterns are natural with respect to phonetic considerations. On the one hand, there is a group of proposals arguing that most, if not all, phonological patterns are phonetically motivated, and/or phonological representations contain detailed phonetic information (Hayes, Kirchner, & Steriade, 2004). On the other hand, some researchers argue that synchronic phonological systems should be completely void of phonetic substance (Reiss, 2018). See Kingston (2019) for the most recent review on this debate. At the observational level, some phonological generalizations do seem to be motivated by phonetic considerations. For instance, voiced stops are generally considered marked compared to voiceless stops, because voiced stops present an aerodynamic challenge (Hayes & Steriade, 2004). In order to maintain glottal vibration, there must be a sufficient trans-glottal air pressure drop; however, stop closure raises intraoral air pressure, and speakers need to expand their oral cavity to accommodate this aerodynamic challenge (Ohala, 1983a). Many languages thus avoid voiced stops in favor of voicless stops, which seems to have its roots in the aerodynamic challenge. Several questions still remain actively debated: (a) whether the influence of phonetics on phonology should be directly encoded in synchronic grammar, or it should be treated as the results of phonetically motivated diachronic changes; (b) whether those phonological patterns that seem “crazy” from the perspective of phonetic naturalness (Bach & Harms, 1972) can be productive; (c) and if so, how we should model natural and unnatural aspects of phonological knowledge.

Just like some phonological generalizations, some sound symbolic connections seem to be grounded in the articulatory or acoustic properties of the sounds at issue. This intuition was already expressed by Sapir (1929: 233), a pioneering work on modern studies of sound symbolism: “the symbolic discriminations run encouragingly parallel to the objective ones based on phonetic considerations.”[4] Jespersen (1922) and Sapir (1929) found that speakers generally judge [a] to be larger than [i], and attributed this observation to either the differences in the degrees of oral aperture or the differences in their resonance frequencies (most likely their F2). To provide another example, voiced obstruents are often considered to be larger than voiceless obstruents by speakers of different languages (Hamano, 1998; Newman, 1933; Shinohara & Kawahara, 2016), and it is not hard to imagine that this image of largeness is grounded in the expansion of the supralaryngeal cavity that is necessitated by the aerodynamic requirement that voiced stops present (Ohala, 1983a). D'Onofrio (2014) shows that segments that involve lip gestures—such as [b] and [u]—tend to be associated with round figures, in the context of studies of what is known as the *bouba-kiki* effect (Ramachandran & Hubbard, 2001). It is again likely that the lip rounding gesture of these sounds leads to the image of roundedness. These sound symbolic patterns do not seem to arise from statistical skews in the lexicon; neither is there a plausible diachronic story in which these relationships arose—they instead appear to emerge in experimental settings using nonce words, suggesting that the influence of phonetic considerations in sound symbolism is synchronically active.

On the other hand, some sound symbolic patterns do not appear to have such clear phonetic motivations. For example, English *gl-* sequences occur in many words that are related to the notion of "light" (e.g., *glitter, glow, gloaming*). However, there are no phonetic reasons to expect that the sequence of *gl-* should be connected to the meaning of "light." It has been shown moreover that this sound–meaning connection is psychologically real; that is, this sound–meaning correspondence cannot be relegated as an accidental connection in the English lexicon (Bergen, 2004).

We also observe cases that are akin to phonetically "crazy rules" (Bach & Harms, 1972) in sound symbolism. For example, in Korean, [ɑ] and [o] are symbolically smaller than [u] and [ʌ] (Garrigues, 1995; Kim, 1977). This pattern flouts an otherwise cross-linguistically common observation that high vowels are generally judged to be smaller than non-high vowels (Sapir, 1929*et seq*), and not only that, to the extent that this sound symbolism has its roots in the different degrees of oral aperture, it runs counter to what we expect from phonetic considerations. In other words, these sound symbolic connections in Korean are phonetically crazy (see also Diffloth, 1994).

In the phonology literature, there is a debate regarding whether phonetically crazy rules can be productive or not (Hayes, Zuraw, Siptár, & Londe, 2009; Kawahara, 2008; Sanders, 2003). Sometimes crazy rules turn out to be non-productive (Sanders, 2003), or there is a learning bias against them (Hayes et al., 2009; Hayes & White, 2013; White, 2014; Wilson, 2006). In this connection, it is interesting that Shinohara and Kawahara (2016) found that given nonce words, Korean speakers judge high vowels to be smaller than low vowels, contrary to what we expect from the lexical patterns. Similar results in which grammatical naturalness, which is demonstrably grounded in phonetic considerations, triumphs unnatural lexical skews are reported in some recent phonological studies (Guilherme, 2019; Jarosz, 2017).

3.2 | Bases of representations—Articulation or acoustics?

Another major debate in phonological theory is the phonetic bases of distinctive features.[5] Jakobson, Fant, and Halle (1952) first formalized distinctive features in terms of acoustic characteristics. On the other hand, the feature set deployed by Chomsky and Halle (1968) was primarily based on articulation. Since then, there has been a heated debate as to whether distinctive features—or phonological representations in general—should be defined based on articulation or acoustics/perception (Kingston, 2007).[6]

A similar debate arises in the studies of sound symbolism. Take the case of [a] being perceived as larger than [i]. Jespersen (1922: 558-559) entertains two hypotheses regarding why [i] is considered to be small, one based on acoustics and one based on articulation: "[t]he reason why the sound [i] comes to be easily associated with small, and [u, o, a] with bigger things, may be to some extent the high pitch of the vowel…; the perception of the small lip aperture in one case and the more open mouth in the other may have also its share in the rise of the idea." Ohala (1994) advocated a general theory of sound symbolic patterns, which is now known as the frequency code hypothesis. In this theory, sounds with high frequency energy (either f0 or F2) evoke images of smallness, since these sounds are generated by a set of small vocal folds (in case of high f0) or a small resonating chamber (in case of high F2).

One argument for the acoustics-based explanation is that in some languages such as Twi, H-tone can represent something small (e.g., [kákrà] “small” vs. [kàkrá] “large”), and there appears to be nothing plausible in the articulation of H-tone that can be connected to the image of smallness (Ohala, 1983b). Another argument for the acoustics-based explanation is that speakers of different languages can order their vowels in terms of how big they sound (Newman, 1933), and the inverse of F2 is almost a perfect predictor of this judgment pattern (Shinohara & Kawahara, 2016).

On the other hand, there are arguments in favor of the articulation-based explanation as well. First, deaf children can detect the sound symbolic values of their own speech and, generally, these patterns are similar to the patterns observed for hearing people (Eberhardt, 1940). Second, the acoustics-based explanation leaves it unclear why F1 does not affect the images of size, at least not as much as F2—[a] has higher F1 than [i], and, therefore, if listeners deduce the sound symbolic values of vowels from their F1, [a] should be judged to be smaller than [i].[7] Third, the connection between labial segments and round shapes (D'Onofrio, 2014) seems to be most straightforwardly explained in terms of articulation, not in terms of acoustics—nothing in the acoustics of labials appears to be “round.”

3.3 | Universality and language-specificity

If sound symbolic patterns have their roots in their phonetic characteristics, one can imagine that sound symbolism is universal, shared across all languages, as we use the same articulatory and perceptual systems. Therefore, the universality of sound symbolic patterns is one topic that is actively discussed. A parallel question—how universal is the phonological system—is one of the central questions in modern linguistics. In both research domains, to the extent that there *are* universals, a deeper question is what level of abstraction is necessary to establish those universals (Akita, 2015).

The universality of sound symbolism has been addressed in two ways. One is a cross-linguistic comparison, often in the form of experimentation. Shinohara and Kawahara (2016), for example, used nonce word stimuli to explore the judgment of size associated with five different vowels, [i], [e], [a], [o], and [u], targeting speakers of Chinese, English, Japanese, and Korean. They found that speakers of all languages judged [a] to be larger than [i], hinting at the universality of this pattern (though see Diffloth, 1994); on the other hand, Japanese speakers judged [o] to be larger than [a], whereas Chinese and Korean speakers showed the opposite pattern, and English speakers did not show a substantial difference between the two vowels. To take another example, the *takete-maluma* effect (Köhler, 1947) has been shown to hold across many languages (Styles & Gawne, 2017), but it fails in Songe (Rogers & Ross, 1975) and Syuba (Styles & Gawne, 2017). Based on a meta-analysis of the previous studies on the *takete-maluma* effect, Styles and Gawne tentatively propose that it fails to hold if the stimuli violate phonotactic restrictions of the target language. See Bremner et al. (2013), Dingemanse, Schuerman, Reinisch, Tufvesson, and Mitterer (2016), Saji, Akita, Kantartzis, Kita, and Imai (2019) and Shih et al. (2019) for further discussion on the universality and language-specificity of sound symbolic patterns.

Another approach is to examine the behavior of pre-verbal infants. As reviewed in Section 4.2, there is now a growing body of work showing that pre-verbal infants are sensitive to cross-linguistically prevalent sound symbolic patterns (Imai, Kita, Nagumo, & Okada, 2008;

Kantartzis, Imai, & Kita, 2011; Maurer, Pathman, & Mondloch, 2006; Ozturk, Krehm, & Vouloumanos, 2013; Peña, Mehler, & Nespor, 2011). On the other hand, some acquisition studies show that not all sound symbolic patterns may be universal. Fort, Weiβ, Martin, and Peperkamp (2013), for example, failed to find an otherwise cross-linguistically robust *bouba-kiki* effect (Ramachandran & Hubbard, 2001) in French infants. Iwasaki, Vinson, and Vigiliocco (2007) found that native speakers of English without any L2 background on Japanese were able to guess the meanings of some sound symbolic, onomatopoetic words in Japanese, but not others, arguing that some sound symbolic patterns are universal while others are not.

Building on these results, Imai and Kita (2014) hypothesize that "young children are sensitive to all possible sound symbolic correspondences that could appear in any language of the world, but only a subset of these correspondences are compatible with the phonological inventory and the existing words in the language the children are learning. As they grow up, the sensitivity to the incompatible correspondences wanes, and adults maintain only the sensitivity to the compatible correspondences." This hypothesis may remind phonologists of Stampe's (1973) proposal that babies are borne with a set of universal, phonetically motivated processes, and as they grow up, they unlearn some of them and acquire language-specific rules.

3.4 | Cumulativity

Another common issue is the question of *cumulativity*. In phonology, this issue is most actively discussed in the context of comparing Optimality Theory (Prince & Smolensky, 2004) with other related constraint-based theories (Pater, 2009; Zuraw & Hayes, 2017): when a structure violates two constraints, do the effects of these two constraints add up? Optimality Theory posits that only the higher-ranked constraint matters, whereas other theories such as Harmonic Grammar, suggest that the expected outcome should be cumulative.

A similar question arises when studying sound symbolism. For instance, some studies report that two instances of the same segment can evoke a stronger image than one instance (Hamano, 2013; Kawahara et al., 2019; Kawahara and Kumagai, to appear; Martin, 1962), which is akin to what Jäger and Rosenbach (2006) refer to as *counting cumulativity*. Moreover, D'Onofrio (2014) shows that in the *bouba-kiki* effect, several phonetic/phonological features matter in determining the perceived roundness/angularity of visual images (e.g., vowel backness, voicing, and place of articulation), and that these effects are cumulative, which would remind phonologists of *ganging-up cumulativity* (Jäger & Rosenbach, 2006). Kumagai and Kawahara (2019) show that Japanese speakers prefer to use low vowels and voiced obstruents for the names of evolved Pokémon characters; while each of these effects is observed independently, when combined in one stimulus, their effects are cumulative. Based on these observations, they present an analysis using MaxEnt Grammar (Goldwater & Johnson, 2003), analogous to the phonological analyses presented in Zuraw and Hayes (2017). See Ahlner and Zlatev (2010) and Thompson and Estes (2011) for other potential cases of cumulative effects in sound symbolism. Kawahara (2020) offers MaxEnt analyses of these cumulative patterns.

3.5 | Positional asymmetry

Yet another common issue is positional asymmetry. It is well-known in the phonology literature that some positions—for example, onsets, stressed syllables, and word-initialsegments—

are privileged in that they can, for example, trigger phonological processes and license a wider range of segments (e.g., Beckman, 1998). Similarly, word-initial voiced obstruents cause stronger sound symbolic image than word-internal voiced obstruents in Japanese (Kawahara, Shinohara, & Uchimoto, 2008) (see also Haynie, Bowern, & LaPalombara, 2014; McGregor, 1996).

Compared to the extensive body of research on positional effects in phonology, this issue is less well examined in the studies of sound symbolism. Exploring positional effects in sound symbolism may offer a novel perspective on this issue, since why and how some positions are privileged is related to the issue of phonetic grounding of phonological patterns (Section 3.1)—positional privileges are often considered as arising from phonetic/psycholinguistic prominence of these positions, but whether this influence should be encoded synchronically or diachronically is much debated (e.g., Barnes, 2002 vs. Smith, 2002). Also, for some positions, it is not clear whether that position is phonologically privileged or not (e.g., word final position). Exploring the positional effects in sound symbolism may help phonologists address this question as well; see for example, McGregor (1996) who explored the sound symbolic values of root-final consonants in Gooniyandi.

4 | ADDITIONAL POTENTIAL BENEFITS

There are additional benefits that phonologists can gain from studying sound symbolism. Due to space limitation, the discussion in this section needs to be brief; see the other overview papers cited in Section 1 as well as the reference cited in this section for further details.

4.1 | Addressing the origin of human languages

First, some researchers propose that mimicking real-world attributes with different types of vocalization contributed to the origin of human languages (Berlin, 2006; Cabrera, 2012; Haiman, 2018; Perlman & Lupyan, 2018; Perniss et al., 2010; Perniss & Vigiliocco, 2014; Ramachandran & Hubbard, 2001). A recent study by Perlman and Lupyan (2018), for example, shows that speakers are able to communicate as many as 30 different meanings by way of iconic vocalizations, well above the chance level; moreover, vocalizations which were judged to be more iconic were learned faster. They conclude that "[t]his newly emerging understanding of iconicity as a widespread property of spoken languages suggests iconicity may also have played an important role in their origin. An intriguing possibility is that many of the now arbitrary words in modern spoken languages may have originated from the innovation of iconic vocalizations." If this hypothesis is on the right track, analyzing sound symbolic patterns might shed light on how human languages may have emerged and evolved.

4.2 | The role of sound symbolism in language acquisition

Some researchers argue that sound symbolism plays a non-trivial role in language acquisition (Asano et al., 2015; Imai et al., 2008; Kantartzis, 2011; Perry, Perlman, Winter, Massaro, & Lupyan, 2018). In the context of first language acquisition, Maurer et al. (2006) demonstrate that 2.5 year old children are sensitive to sound symbolic associations. Even more strikingly, Peña et al. (2011) show that this sensitivity is detectable in 4 month old neonates. These observations have led to the general hypothesis that sound symbolism guides the first language acquisition process (Imai & Kita, 2014; see §3.3).

A partial support of this theory comes from the observation that Japanese caretakers use more sound symbolic, onomatopoetic words to infants and children than to adults (Fernald & Morikawa, 1993). Sound symbolism may thus provide a partial answer to the question of why human children acquire languages so quickly—a fundamental question that theoretical linguists attempt to answer. In addition, it has been observed that, in the context of L2 acquisition too, those items that follow sound symbolic principles are easier to learn and more frequently used by language learners (Kunihara, 1971; Nygaard, Cook, & Namy, 2009).

4.3 | Tighter connections with cognitive science

Sound symbolism is now considered as a specific instance of general cross-modal, synesthetic correspondences, in which sensation in one modality has correspondences with sensation in another modality (Bankieris & Simner, 2015; Cuskley & Kirby, 2013; Sidhu & Pexman, 2018; Spence, 2011). There has been a growing body of interest in these cross-modal perception patterns in cognitive science, exploring not only the relationship between sounds and meanings, but also the relationships between sounds and cognitive patterns in other modalities (such as vision and taste). Therefore, engaging with linguistic analyses of sound symbolism has the potential to facilitate more extensive interdisciplinary communication between phonologists and cognitive scientists. Two questions addressed in this research that are of particular interest to theoretical linguists are: (a) are sound symbolism/synesthetic connections innate or acquired, and (b) are sound symbolic connections deducible to a domain-general synesthetic mechanism? (Akita, 2015: §4.3).

4.4 | Sound symbolism and branding

There is an increasing body of work that seeks to deploy sound symbolism in the context of marketing. Their general finding is that there are sounds that are "suitable" to convey particular images for brand products; such names that make sound symbolic sense are judged to be better, and possibly better remembered, by potential customers (Bolts, Mangigian, & Allen, 2016; Coulter & Coulter, 2010; Jurafsky, 2014; Klink, 2000; Peterson & Ross, 1972; Yorkston & Menon, 2004). This line of research has opened up a new domain of interdisciplinary research.

4.5 | Popularizing linguistics and application to pedagogy

Finally, studying sound symbolism may help us popularize linguistics, and relatedly, can be useful for teaching. One example that instantiates these points is a study of sound symbolic patterns in Pokémon names (Kawahara, Noto, & Kumagai, 2018), which has shown, for example, that Pokémon characters' weight positively correlates with the number of voiced obstruents contained in their names. This result was featured in various popular magazines in Japan. Stephanie Shih, who followed up the original study with a much wider range of languages (Shih et al., 2019), was featured in a radio show to talk about this project. Being able to show the general public that analytical tools that phonologists use—such as voiced obstruents—are useful for analyzing Pokémon names can be appealing. This feature of sound symbolism also often attracts undergraduate students' interests as well, since the targets of the analyses include "fun materials" that they are already interested in (e.g., Pokémon). See Kawahara (2019) and MacKenzie (2018) for reports of the usefulness of using onomastics, including sound symbolism, in undergraduate education.

5 | CONCLUSION

Studying sound symbolic patterns can be interesting and informative for theoretical phonologists. To recap, there are some alternation patterns that seem to be motivated by sound symbolic principles, which can interact with other phonological considerations. There are many common interests shared by theoretical phonologists and those who work on sound symbolism, and we can mutually inform one another. There are various additional benefits that phonologists can gain from working on sound symbolism. Exploration of sound symbolism may contribute to the further development of the field by attracting interests from students and researchers in other fields.

ACKNOWLEDGEMENTS

I am grateful to my collaborators, especially Gakuji Kumagai and Kazuko Shinohara, for thinking with me about what studying sound symbolism as a theoretical linguist means. I thank Mark Dingemanse, Donna Erickson, Kazuko Shinohara, the associate editor Jie Zhang and two anonymous reviewers for comments on previous versions of this paper. This project is supported by the Japan Society for the Promotion of Science grant #17K13448. Remaining errors are mine.

CONFLICT OF INTEREST

The authors declares no potential conflict of interest.

ORCID

Shigeto Kawahara https://orcid.org/0000-0001-9984-0576

ENDNOTES

[1]I acknowledge that there are general disagreements regarding what the domain of phonological inquiry should include. de Lacy (2009), for example, excludes loanword adaptation patterns and diachronic changes from topics of phonological inquiry, while both of these receive extensive attention from other phonologists. Some researchers are more willing to accept "external evidence" (Churma, 1979)—for example, language game patterns or rhyming patterns—for phonological argumentation than others (Ohala, 1986).

[2]I do not have quantitive evidence for this claim, but Alderete & Kochetov (2017: 731) state that "it is fair to say that sound symbolism has never found a natural place in generative grammar." I note that there have been phonological analyses of alternations that are demonstrably driven by sound symbolic principles (see Section 2.1). There are also a few monumental studies in the 90s that examined sound symbolic connections from a linguistic point of view, which include Hinton, Nichols, and Ohala (1994) and Hamano (1998). Ideophones have long been considered as exceptional to the Saussuren theorem of arbitrariness; their linguistic behaviors have attracted attention from theoretical linguists in the 70s and 80s (Dingemanse, 2018: §5.2), and recently, there have been renewed interests in the linguistic analyses of ideophones (Akita, 2019; Akita & Dingemanse, 2019).

[3]This is not to say that all sound symbolic patterns can be defined based on distinctive features (Akita, 2015). However, not all phonological alternations can be neatly captured in terms of distinctive features either (Mielke, 2008). This complexity may instantiate a parallel between phonological alternations and sound symbolic patterns (see Section 3).

[4]This observation actually goes back to Socrates, who in *Cratylus* (427) discusses relationships between sound symbolic meanings and phonetic properties of the sounds at issue.

[5]A similar debate exists in the phonetics literature regarding whether phonetic targets should be defined articulatorily or acoustically, and also regarding whether the object of speech perception is articulation or acoustics/audition (Kingston, 2007).

[6]Setting aside the view that phonological representations should completely lack phonetic substance (Reiss, 2018).

[7]Knoeferle, Li, Maggioni, and Spence (2017) show that stimuli with higher F1 can lead to higher size rating.

REFERENCES

Ahlner, F., & Zlatev, J. (2010). Cross-modal iconicity: A cognitive semiotic approach to sound symbolism. *Sign Sytems Studies*, *38*(1/4), 298–348.

Akita, K. (2015). Sound symbolism. In J.-O. Östman & J. Verschueren (Eds.), *Handbook of pragmatics, installment 2015*. Amsterdam and Philadelphia: John Benjamins.

Akita, K. (2019). Ideophones. In M. Aronoff (Ed.), *Oxford bibiliographies in linguistics*. New York, NY: Oxford University Press.

Akita, K., & Dingemanse, M. (2019). Ideophones (mimetics, expressives). In M. Aronoff (Ed.), *Oxford research encyclopedia of linguistics*. Oxford, England: Oxford University Press.

Albright, A. (2009). Feature-based generalisation as a source of gradient acceptability. *Phonology*, *26*(1), 9–41.

Alderete, J., & Kochetov, A. (2017). Integrating sound symbolism with core grammar: The case of expressive palatalization. *Language*, *93*, 731–766.

Asano, M., Imai, M., Kita, S., Kitaji, K., Okada, H., & Thierry, G. (2015). Sound symbolism scaffolds language development in preverbal infants. *Cortex*, *63*, 196–205.

Bach, E., & Harms, R. (1972). How do languages get crazy rules? In R. Stockwell & R. Macaulay (Eds.), *Linguistic change and generative theory* (pp. 1–21). Bloomington, IN: Indiana University Press.

Bankieris, K., & Simner, J. (2015). What is the link between synaesthesia and sound symbolism?*Cognition*, *136*, 186–195.

Barnes, J. (2002). Positional neutralization: A phonologization approach to typological patterns. Doctoral dissertation, University of California, Berkeley.

Beckman, J. (1998). Positional faithfulness. Doctoral dissertation, University of Massachusetts, Amherst.

Bergen, B. K. (2004). The psychological reality of phonaesthemes. *Language*, *80*, 290–311.

Berlin, B. (2006). The first congress of ethonozoological nomenclature. *Journal of Royal Anthropological Institution*, *12*, 23–44.

Bhat, D. N. S. (1978). A general study of palatalization. In J. Greenberg (Ed.), *Universals of human language: Phonology* (Vol. 2, pp. 47–92). Stanford, CA: Stanford University Press.

Boersma, P., & Hayes, B. (2001). Empirical tests of the Gradual Learning Algorithm. *Linguistic Inquiry*, *32*, 45–86.

Bolts, M. G., Mangigian, M. G., & Allen, M. B. (2016). Phonetic symbolism and memory for advertisement. *Applied Cognitive Psychology*, *30*, 1088–1092.

Bremner, A. J., Caparos, S., Davidoff, J., de Fockert, J., Linnell, K. J., & Spence, C. (2013). "Bouba" and "Kiki" in Namibia? A remote culture make similar shape-sound matches, but different shape-taste matches to westerners. *Cognition*, *126*, 165–172.

Cabrera, J. C. M. (2012). The role of sound symbolism in protolanguage: Some linguisitic and archaeological speculations. *Theoria et Historia Scientiarum*, *9*, 115–130.

Chomsky, N., & Halle, M. (1968). *The sound pattern of English*. New York, NY: Harper and Row.

Churma, D. (1979). Arguments from external evidence in phonology. Doctoral dissertation, Ohio State University.

Coetzee, A. W., & Pater, J. (2011). The place of variation in phonological theory. In J. A. Goldsmith, J. Riggle, & A. Yu (Eds.), *The handbook of phonological theory* (2nd ed., pp. 401–431). Oxford: Blackwell-Wiley.

Coulter, S. K., & Coulter, R. A. (2010). Small sounds, big deals: Phonetic symbolism effects in pricing. *Journal of Consumer Research*, *37*(2), 315–328.

Cuskley, C., & Kirby, S. (2013). Synesthesia, cross-modality, and language evolution. In J. Simner & E. Hubbard (Eds.), *Oxford handbook of synesthesia*. Oxford: Oxford University Press.

D'Onofrio, A. (2014). Phonetic detail and dimensionality in sound-shape correspondences: Refining the *bouba-kiki* paradigm. *Language and Speech*, *57*(3), 367–393.

de Lacy, P. (2009). Phonological evidence. In S. Parker (Ed.), *Phonological argumentation: Essays on evidence and motivation* (pp. 43–77). London, England: Equinox.

Diffloth, G. (1994). i: *big*, a: *small*. In L. Hinton, J. Nichols, & J. J. Ohala (Eds.), *Sound symbolism* (pp. 107–114). Cambridge, England: Cambridge University Press.

Dingemanse, M. (2018). Redrawing the margins of language: Lessons from research on ideophones. *Glossa*, *3*(1), 4. https://doi.org/10.5334/gjgl.444

Dingemanse, M., Blasi, D. E., Lupyan, G., Christiansen, M. H., & Monaghan, P. (2015). Arbitrariness, iconicity and systematicity in language. *Trends in Cognitive Sciences*, *19*(10), 603–615.

Dingemanse, M., Schuerman, W., Reinisch, E., Tufvesson, S., & Mitterer, H. (2016). What sound symbolism can and cannot do: Testing the iconicity of ideophones from five languages. *Langauge*, *92*(2), e117–e133.

Eberhardt, M. (1940). A study of phonetic symbolism of deaf children. *Psychological Monograph*, *52*, 23–42.

Fernald, A., & Morikawa, H. (1993). Common themes and cultural variations in Japanese and American mothers' speech to infants. *Child Development*, *64*, 637–656.

Finley, S., & Bedecker, W. (2009). Artificial language learning and feature-based generalization. *Journal of Memory and Language*, *61*, 423–437.

Fort, M., Weiβ, A., Martin, A., and Peperkamp, S. (2013). Looking for the bouba-kiki effect in prelexical infants. Paper presented at: Proceedings of the international conference on auditory-visual speech processing, pp. 71–76.

Garrigues, S. L. (1995). Mimetic parallels in Korean and Japanese. *Studies in Language*, *19*, 359–398.

Goldwater, S. and Johnson, M. (2003). Learning OT constraint rankings using a maximum entropy model. Proceedings of the workshop on variation within Optimality Theory, pp. 111–120.

Guilherme, D. G. (2019). When lexical statistics and the grammar conflict: Learning and repairing weight effects on stress. *Language*, *95*(4), 612–641.

Haiman, J. (2018). *Ideophones and the evolution of language*. Cambridge, England: Cambridge University Press.

Hamano, S. (1998). *The sound-symbolic system of Japanese*. Stanford, CA: CSLI Publications.

Hamano, S. (2013). Hoogen-ni okeru giongo-gitaigo-notaikeiteki-kenkyuu-no igi. In K. Shinohara & R. Uno (Eds.), *Chikazuku Oto-to Imi: Onomatope Kenkyuu-no Shatei*. Tokyo, Japan: Hitsuzi Syobo.

Hayes, B., Kirchner, R., & Steriade, D. (Eds.). (2004). *Phonetically based phonology*. Cambridge, England: Cambridge University Press.

Hayes, B., & Londe, Z. (2006). Stochastic phonological knowledge: The case of Hungarian vowel harmony. *Phonology*, *23*, 59–104.

Hayes, B., & Steriade, D. (2004). Introduction: The phonetic bases of phonological markedness. In B. Hayes, R. Kirchner, & D. Steriade (Eds.), *Phonetically based phonology* (pp. 1–33). Cambridge, England: Cambridge University Press.

Hayes, B., & White, J. (2013). Phonological naturalness and phonotactic learning. *Linguistic Inquiry*, *44*, 45–75.

Hayes, B., Zuraw, K., Siptár, P., & Londe, Z. (2009). Natural and unnatural constraints in Hungarian vowel harmony. *Language*, *85*(4), 822–863.

Haynie, H., Bowern, C., & LaPalombara, H. (2014). Sound symbolism in the languages of Australia. *PLoS One*, *9*(4), e92852. https://doi.org/10.1371/journal.pone.0092852

Hinton, L., Nichols, J., & Ohala, J. (1994). *Sound symbolism*. Cambridge, England: Cambridge University Press.

Hockett, C. (1959). Animal "languages" and human language. *Human Biology*, *31*, 32–39.

Imai, M., & Kita, S. (2014). The sound symbolism bootstrapping hypothesis for language acquisition and language evolution. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *369*(1651), 20130298.

Imai, M., Kita, S., Nagumo, M., & Okada, H. (2008). Sound symbolism facilitates early verb learning. *Cognition*, *109*, 54–65.

Ito, J., & Mester, A. (1995). Japanese phonology. In J. Goldsmith (Ed.), *The handbook of phonological theory* (pp. 817–838). Oxford: Blackwell.

Iwasaki, N., Vinson, D. P., & Vigiliocco, G. (2007). What do English speakers know about *gera-gera* and *yota-yota*? A cross-linguistic investigation of mimetic words for laughing and walking. *Japanese Language Education around the Globe*, *17*, 53–78.

Jäger, G., & Rosenbach, A. (2006). The winner takes it all—almost: Cumulativity in grammatical variation. *Linguistics*, *44*(5), 937–971.

Jakobson, R. (1960). *Language in literature*. Cambridge, England: Harvard University Press.

Jakobson, R., Fant, G., & Halle, M. (1952). *Preliminaries to speech analysis*. MIT Acoustics Laboratory: Technical report.

Jang, H. (2019). *How cute do I sound?: The iconic function of segmental alternation in Korean baby-talk register*. Aegyo. Ms: University of Southern California.

Jarosz, G. (2017). Defying the stimulus: Acquisition of complex onsets in Polish. *Phonology*, *34*(2), 269–298.

Jespersen, O. (1922). Symbolic value of the vowel *i*. In *Linguistica: Selected papers in English, French and German, volume 1* (pp. 283–303). Copenhagen: Levin and Munksgaard.

Jurafsky, D. (2014). *The language of food: A linguist reads the menu*. New York, NY: W. W. Norton & Company.

Kantartzis, K. (2011). Children and adults' understanding and use of sound symbolism in novel words. Doctoral dissertation, University of Birmingham.

Kantartzis, K., Imai, M., & Kita, S. (2011). Japanese sound symbolism facilitates word learning in English-speaking children. *Cognitive Science*, *35*(3), 575–586.

Kawahara, S. (2008). Phonetic naturalness and unnaturalness in Japanese loanword phonology. *Journal of East Asian Linguistics*, *17*(4), 317–330.

Kawahara, S. (2018). Phonology and the orthography: The orthographic characterization of rendaku and Lyman's law. *Glossa*, *3*(1), 10. https://doi.org/10.5334/gjgl.368

Kawahara, S. (2019). Teaching phonetics through sound symbolism. Proceedings of ISAPh.

Kawahara, S. (2020). *Cumulative effects in sound symbolism*, Tokyo: Ms. Keio University.

Kawahara, S., Katsuda, H., & Kumagai, G. (2019). Accounting for the stochastic nature of sound symbolism using Maximum Entropy Model. *Open Linguistics*, *5*, 109–120.

Kawahara, S., & Kumagai, G. (to appear). What voiced obstruents symbolically represent in Japanese: Evidence from the Pokémon universe. *Journal of Japanese Linguistics*.

Kawahara, S., Noto, A., & Kumagai, G. (2018). Sound symbolic patterns in Pokémon names. *Phonetica*, *75*(3), 219–244.

Kawahara, S., Shinohara, K., and Uchimoto, Y. (2008). A positional effect in sound symbolism: An experimental study. Paper presented at: Proceedings of the Japan cognitive linguistics association 8, pp. 417–427. JCLA, Tokyo.

Kim, K.-O. (1977). Sound symbolism in Korean. *Journal of Linguistics*, *13*, 67–75.

Kingston, J. (2007). The phonetics-phonology interface. In P. de Lacy (Ed.), *The Cambridge handbook of phonology* (pp. 401–434). Cambridge, England: Cambridge University Press.

Kingston, J. (2019). The interface between phonetics and phonology. In W. F. Katz & P. F. Assmann (Eds.), *The Routledge handbook of phonetics* (pp. 359–400). London, England and New York, NY: Routledge.

Klink, R. R. (2000). Creating brand names with meaning: The use of sound symbolism. *Marketing Letters*, *11* (1), 5–20.

Knoeferle, K., Li, J., Maggioni, E., & Spence, C. (2017). What drives sound symbolism? Different acoustic cues underlie sound-size and sound-shape mappings. *Scientific Reports*, *7*, 5562.

Köhler, W. (1947). *Gestalt psychology: An introduction to new concepts in modern psychology*. New York, NY: Liveright.

Kumagai, G. (2019). A sound-symbolic alternation to express cuteness and the orthographic Lyman's law in Japanese. *Journal of Japanese Linguistics*, *35*(1), 39–74.

Kumagai, G., & Kawahara, S. (2019). Effects of vowels and voiced obstruents on Pokémon names: Experimental and theoretical approaches [in Japanese]. *Journal of the Linguistic Society of Japan*, *155*, 65–99.

Kumagai, G., & Kawahara, S. (2020). How abstract is sound symbolism? Labiality and diaper names in Japanese [in Japanese]. *Journal of the Linguistic Society of Japan*, 157.

Kunihara, S. (1971). Effects of the expressive force on phonetic symbolism. *Journal of Verbal Learning and Verbal Behavior*, *10*, 427–429.

Lockwood, G., & Dingemanse, M. (2015). Iconicity in the lab: A review of behavioral, developmental, and neuroimaging research into sound-symbolism. *Frontiers in Psychology*, *6*, 1246. https://doi.org/10.3389/fpsyg.2015.01246

Lupyan, G., & Winter, B. (2018). Language is more abstract than you think, or, why aren't languages more iconic? *Proceedings of Royal Society B*, *373*, 20170137.

MacKenzie, L. (2018). What's in a name? Teaching linguistics using onomastic data. *Language [Teaching Linguistics]*, *94*, e1–e18.

Martin, S. (1962). Phonetic symbolism in Korean. In N. Poppe (Ed.), *American studies in Uralic and Altaic linguistics*. Bloomington, IN: Indiana University Press.

Maurer, D., Pathman, T., & Mondloch, C. J. (2006). The shape of boubas: Sound-shape correspondences in toddlers and adults. *Developmental Science*, *9*, 316–322.

McCarthy, J. J. (1983). Phonological features and morphological structure. In J. Richardson, M. Marks, & A. Chukerman (Eds.), *Proceedings from the parasesion on the interplay of phonology, morphology and syntax* (pp. 135–161). Chicago: CLS.

McGregor, W. (1996). Sound symbolism in Gooniyandi, a language of Western Australia. *Word*, *47*(3), 339–364.

Mielke, J. (2008). *The emergence of distinctive features*. Oxford: Oxford University Press.

Newman, S. (1933). Further experiments on phonetic symbolism. *American Journal of Psychology*, *45*, 53–75.

Nobile, L. (2015). Phonemes as images: An experimental inquiry into shape-sound symbolism applied to the distinctive features of French. In M. Hiraga, W. Herlofsky, K. Shinohara, & K. Akita (Eds.), *Iconicity: East meets west* (pp. 71–91). Amsterdam: John Benjamins.

Nuckolls, J. B. (1999). The case for sound symbolism. *Annual Review of Anthropology*, *28*, 225–252.

Nygaard, L. C., Cook, A. E., & Namy, L. L. (2009). Sound to meaning correspondance facilitates word learning. *Cognition*, *112*, 181–186.

Ohala, J. J. (1983a). The origin of sound patterns in vocal tract constraints. In P. MacNeilage (Ed.), *The production of speech* (pp. 189–216). New York, NY: Springer-Verlag.

Ohala, J. J. (1983b). The phonological end justifies any means. In S. Hattori, & K. Inoue (Eds.), Proceedings of the 13th international congress of linguists, pp. 232–243. Sanseido, Tokyo.

Ohala, J. J. (1986). Consumer's guide to evidence in phonology. *Phonology*, *3*, 3–26.

Ohala, J. J. (1994). The frequency code underlies the sound symbolic use of voice pitch. In L. Hinton, J. Nichols, & J. J. Ohala (Eds.), *Sound symbolism* (pp. 325–347). Cambridge, England: Cambridge University Press.

Ozturk, O., Krehm, M., & Vouloumanos, A. (2013). Sound symbolism in infancy: Evidence for sound-shape-cross-modal correspondences in 4-month-olds. *Journal of Experimental Child Psychology*, *14*(2), 173–186.

Pater, J. (2009). Weighted constraints in generative linguistics. *Cognitive Science*, *33*, 999–1035.

Peña, M., Mehler, J., & Nespor, M. (2011). The role of audiovisual processing in early conceptual development. *Psychological Science*, *22*(11), 1419–1421.

Perlman, M., & Lupyan, G. (2018). People can create iconic vocalizations to communicate various meanings to naïve listeners. *Scientific Reports*, *8*(1), 26–34.

Perniss, P., Thompson, R. L., & Vigiliocco, G. (2010). Iconicity as a general property of language: Evidence from spoken and signed languages. *Frontiers in Psychology*, *1*, 1–15. https://doi.org/10.3389/fpsyg.2010.00227

Perniss, P., & Vigiliocco, G. (2014). The bridge of iconicity: From a world of experience to the experiment of language. *Philosophical Transactions of the Royal Society B*, *369*, 20130300.

Perry, L. K., Perlman, M., Winter, B., Massaro, D. W., & Lupyan, G. (2018). Iconicity in the speech of children and adults. *Developmental Science*, *21*(3), e12572.

Peterson, R. A., & Ross, I. (1972). How to name new brand names. *Journal of Advertising Research*, *12*(6), 29–34.

Pierrehumbert, J. B. (2001). Stochastic phonology. *GLOT*, *5*, 1–13.

Prince, A., & Smolensky, P. (2004). *Optimality theory: Constraint interaction in generative grammar*. Malden and Oxford: Blackwell.

Ramachandran, V. S., & Hubbard, E. M. (2001). Synesthesia–a window into perception, thought, and language. *Journal of Consciousness Studies*, *8*(12), 3–34.

Reiss, C. (2018). Substance free phonology. In S. J. Hannahs & A. Bosch (Eds.), *Routledge handbook of phonological theory* (pp. 425–452). London, England and New York, NY: Routledge.

Rogers, S. K., & Ross, A. S. (1975). A cross-cultural test of the maluma-takete phenomenon. *Perception*, *4*, 105–106.

Saji, N., Akita, K., Kantartzis, K., Kita, S., & Imai, M. (2019). Cross-linguistically shared and language-specific sound symbolism in novel words elicited by locomotion videos in Japanese and English. *PLoS One*, *14*(7), e0218707. https://doi.org/10.1371/journal.pone.0218707

Sanders, N. (2003). Opacity and sound change in the Polish lexicon. Doctoral dissertation, University of California, Santa Cruz.

Sapir, E. (1929). A study in phonetic symbolism. *Journal of Experimental Psychology*, *12*, 225–239.

Saussure, F. (1916). *Cours de linguistique générale*. Paris: Payot.

Schmidtke, D. S., Conrad, M., & Jacobs, A. M. (2014). Phonological iconicity. *Frontiers in Psychology*, *5*, 80. https://doi.org/10.3389/fpsyg.2014.00080

Schütze, C. (1996). *The empirical base of linguistics: Grammaticality judgments and linguistic methodology*. Chicago, IL: University of Chicago Press.

Shih, S. S., Ackerman, J., Hermalin, N., Inkelas, S., Jang, H., Johnson, J., Kavitskaya, D., Kawahara, S., Oh, M., Starr, R. L., and Yu, A. (2019). Cross-linguistic and language-specific sound symbolism: Pokémonastics. Ms. University of Southern California, University of California, Merced, University of California, Berkeley, Keio University, National University of Singapore and University of Chicago.

Shinohara, K. and Kawahara, S. (2016). A cross-linguistic study of sound symbolism: The images of size. Paper presented at: Proceedings of the Thirty Sixth Annual Meeting of the Berkeley Linguistics Society. pp. 396–410. Berkeley Linguistics Society, Berkeley.

Sidhu, D., & Pexman, P. M. (2018). Five mechanisms of sound symbolic association. *Psychonomic Bulletin & Review*, *25*(5), 1619–1643.

Sidhu, D. M., Deschamps, K., Bourdage, J. S., & Pexman, P. M. (2019). Does the name say it all? Investigating phoneme-personality sound symbolism in first names. *Journal of Experimental Psychology: General*, *148*(9), 1595–1614.

Smith, J. (2002). Phonological augmentation in prominent positions. Doctoral dissertation, University of Massachusetts, Amherst.

Spence, C. (2011). Crossmodal correspondences: A tutorial review. *Attention, Perception & Psychophysics*, *73*(4), 971–995.

Stampe, D. (1973). A Dissertation on Natural Phonology. Doctoral dissertation, University of Chicago. Published by Garland, New York, 1979.

Styles, S. J., & Gawne, L. (2017). When does maluma/takete fail? Two key failures and a meta-analysis suggest that phonology and phonotactics matter. *Perception*, *8*(4), 1–17.

Svantesson, J.-O. (2017). Sound symbolism: The role of word sound in meaning. *Wiley Interdisciplinary Reviews: Cognitive Science*, e01441.*8*,

Thompson, P. D., & Estes, Z. (2011). Sound symbolic naming of novel objects is a graded function. *Quarterly Journal of Experimental Psychology*, *64*(12), 2392–2404.

Ultan, R. (1978). Size-sound symbolism. In J. Greenberg (Ed.), *Universals of human language II: Phonology* (pp. 525–568). Stanford, CA: Stanford University Press.

White, J. (2014). Evidence for a learning bias against saltatory phonological alternations. *Cognition*, *130*(1), 96–115.

Wilson, C. (2006). Learning phonology with substantive bias: An experimental and computational study of velar palatalization. *Cognitive Science*, *30*(5), 945–982.

Yorkston, E., & Menon, G. (2004). A sound idea: Phonetic effects of brand names on consumer judgments. *Journal of Consumer Research*, *31*, 43–51.

Zuraw, K., & Hayes, B. (2017). Intersecting constraint families: An argument for Harmonic Hrammar. *Language*, *93*, 497–548.

AUTHOR BIOGRAPHY

Dr. Shigeto Kawahara is an associate professor in the Keio Institute of Cultural and Linguistic Studies. He obtained his PhD in linguistics in 2007 from the University of Massachusetts Amherst, and taught phonetics and phonology at the University of Georgia and Rutgers University, before moving to Keio University. He is an author of three introductory phonetics books written in Japanese, and has published many research articles in various journals, including Journal of Phonetics, Language, Natural Language and Linguistic Theory, Language and Speech, Phonetica and Phonology. His research interests include phonetics, phonology and their interface, as well as sound symbolism. He takes both experimental and theoretical approaches to explore the nature of linguistic knowledge.

How to cite this article: Kawahara S. Sound symbolism and theoretical phonology. *Lang Linguist Compass*. 2020;e12372. https://doi.org/10.1111/lnc3.12372